

Name: _____

Date: _____

SEQUENCES, (GEOMETRIC SEQUENCES)

LO: I can identify, generate, and find the nth term of geometric sequences.

What is a geometric sequence?

A **geometric sequence** has a constant **common ratio** (r) between consecutive terms. Each term is found by multiplying the previous term by r .

Formula: The n th term is $u_n = ar^{n-1}$, where a is the first term and r is the common ratio.

Quick examples

●	2, 6, 18, 54, ... ($r = 3$)	multiply by 3
●	100, 50, 25, 12.5, ... ($r = 0.5$)	multiply by $\frac{1}{2}$
●	$u_n = 3 \times 2^{n-1}$ gives 3, 6, 12, 24, ...	$a=3, r=2$
●	2, 4, 6, 8, ... is geometric	that is arithmetic (+2)
●	$u_n = ar^n$ (should be $n-1$)	exponent is $n-1$

Key idea

Divide consecutive terms to find r . The n th term formula is $u_n = ar^{n-1}$.

$$u_n = \textcircled{a} \times \textcircled{r^{n-1}}$$

a = first term, r = common ratio, n = term number

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - finding the common ratio

Find the common ratio of 5, 15, 45, 135, ...

$$r = \text{second} \div \text{first} = 15 \div 5 = 3$$

$$\text{Check: } 45 \div 15 = 3, 135 \div 45 = 3$$

$$\text{Common ratio } r = 3$$

$$\boxed{r = 3}$$

Divide any term by the previous term to find r .

Example 2 - finding the nth term

Find the 8th term of 2, 6, 18, 54, ...

$$a = 2, r = 3$$

$$u_8 = 2 \times 3^7$$

$$u_8 = 2 \times 2187 = 4374$$

$$\boxed{u_8 = 4374}$$

Use $u_n = ar^{n-1}$ with careful power calculation.

Example 3 - fractional ratio

Find the 5th term of 64, 32, 16, 8, ...

$$a = 64, r = 32 \div 64 = \frac{1}{2}$$

$$u_5 = 64 \times \left(\frac{1}{2}\right)^4$$

$$u_5 = 64 \times \frac{1}{16} = 4$$

$$\boxed{u_5 = 4}$$

When r is a fraction, terms decrease.

Example 4 - writing the nth term formula

Write the n th term formula for 3, 12, 48, 192, ...

$$a = 3, r = 12 \div 3 = 4$$

$$u_n = 3 \times 4^{n-1}$$

$$\text{Check: } u_3 = 3 \times 16 = 48$$

$$\boxed{u_n = 3 \times 4^{n-1}}$$

Substitute a and r into the formula $u_n = ar^{n-1}$.

YOUR TURN – PRACTICE (deliberate practice)

1. Finding the common ratio

Find the common ratio.

- a) 2, 10, 50, 250, ...
- b) 81, 27, 9, 3, ...
- c) 4, -8, 16, -32, ...
- d) 1000, 100, 10, 1, ...

2. Continuing sequences

Find the next two terms.

- a) 3, 6, 12, 24, ...
- b) 5, 20, 80, 320, ...
- c) 256, 128, 64, 32, ...
- d) 1, -3, 9, -27, ...

3. Finding specific terms

Use $u_n = ar^{n-1}$ to find the stated term.

- a) $a = 2, r = 3$. Find u_6 .
- b) $a = 5, r = 2$. Find u_8 .
- c) $a = 100, r = 0.5$. Find u_5 .
- d) $a = 1, r = 10$. Find u_4 .

4. Writing nth term formulas

Write the nth term formula for each sequence.

- a) 4, 12, 36, 108, ...
- b) 7, 14, 28, 56, ...
- c) 500, 100, 20, 4, ...
- d) 1, -2, 4, -8, ...

5. Mixed geometric problems

Answer each question.

- a) 3rd term = 12, $r = 2$. Find the 1st term.
- b) 1st term = 5, 4th term = 40. Find r .
- c) Sequence: 2, 6, 18, ... Which term is 1458?
- d) $a = 1000, r = 0.8$. Find u_3 .
- e) Sum of first 4 terms of 1, 2, 4, 8, ...

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) In a geometric sequence, you multiply by the same ratio each time ☐

Correct answer: _____

Reason: _____

b) 2, 4, 6, 8 is a geometric sequence ☐

Correct answer: _____

Reason: _____

c) The nth term is $u_n = ar^{n-1}$ ☐

Correct answer: _____

Reason: _____

d) The nth term is $u_n = ar^n$ ☐

Correct answer: _____

Reason: _____

e) A negative ratio causes terms to alternate in sign ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A geometric sequence has $u_2 = 12$ and $u_5 = 324$. Find the common ratio and the first term.

Expression: _____

Simplified: _____

b) A ball is dropped from 2 metres. Each bounce reaches $\frac{3}{4}$ of the previous height. After how many bounces is the ball below 50 cm? Find the total vertical distance travelled after 4 bounces.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

